

AML Transaction Monitoring on GCP

Project Report · Google Cloud (BigQuery, Pub/Sub, Cloud Storage, Cloud Functions) · Apache Airflow / Cloud Composer · Python, SQL

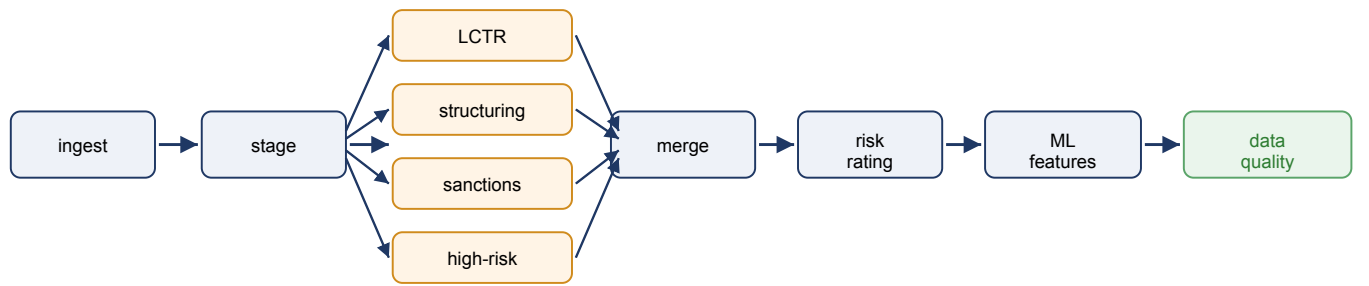
1. Purpose

Banks must detect money laundering hidden inside large volumes of transactions and report suspicious activity to the regulator (FINTRAC). This project implements an automated AML transaction-monitoring platform that ingests transactions, runs FINTRAC-style detection rules, scores each customer's risk, and prepares a training-ready dataset for an AML model — all guarded by data-quality checks.

2. Architecture — three ingestion modes

Mode	Trigger	Tech	Use case
Batch	scheduled / manual	Airflow + BigQuery	daily detection, structuring, risk rating
Streaming	each Pub/Sub message	Pub/Sub + Dataflow (Beam)	real-time per-transaction screening
Event-driven	file lands in GCS	Cloud Function	auto-ingest upstream feeds

Batch DAG flow:



3. Detection rules

Rule	Logic	Severity
LCTR	single cash transaction $\geq 10,000$ (FINTRAC large-cash)	HIGH
Structuring	≥ 3 sub-threshold cash txns by one customer in a day, sum $> 10,000$	HIGH
Sanctions / name screening	counterparty matches the watchlist	CRITICAL
High-risk geography	counterparty in a high-risk country	MEDIUM

Alerts roll up into a **Customer Risk Rating** (severity-weighted score → LOW / MEDIUM / HIGH) and an **ml_features** table provides a labelled, training-ready dataset for an AML model.

4. Results (latest run)

Input: **1,848 transactions** · 300 customers · 5 days.

Stage	Table(s)	Rows
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Raw / reference	raw_transactions / watchlist / high_risk_countries	1,848 / 4 / 4
Staging	stg_transactions	1,848
Detection	lctr / structuring / sanctions / high_risk	8 / 6 / 7 / 7
Consolidated	alerts	28
Scoring	customer_risk_rating	27 (HIGH 7 · MEDIUM 14 · LOW 6)
ML	ml_features	298
Streaming (real-time)	alerts_realtime	29
Event-driven	raw_landing	408

Data-quality gate: passed — alerts reconcile to the sum of the rule tables, and the risk-rating customer count matches distinct alerted customers.

5. Design decisions

- **Right tool per trigger** — batch for day-long patterns (structuring); streaming for instant red flags (sanctions); event-driven for unpredictable file feeds.
- **Idempotency** — every task uses CREATE OR REPLACE / truncate-reload, so retries and backfills never duplicate data.
- **Quality as a gate** — the final task fails the DAG run if reconciliation breaks, so wrong numbers never reach a regulator.
- **Layered model** — raw → staging → detection → scoring → ML keeps concerns isolated and changes contained.
- **Least-privilege & lineage** — dedicated identity; transformations in version control for traceability.

6. Cost

Local Airflow (free) orchestrates serverless BigQuery jobs (pennies, within free tier). No always-on Cloud Composer environment; the Cloud Function scales to zero when idle.